

Zihao Wu

Education	Harvard University	Cambridge, MA
	Ph.D. Candidate. Advisor: Daniel J. Eisenstein	2023 – Expected 2028
	Peking University	Beijing, CN
	B.S. in Astrophysics (GPA: 3.82, Rank 1/38)	2019 – 2023
Research Interests	1. Galaxy Formation in the Early Universe 2. Galaxy Large-scale Structures and Proto-clusters 3. Observations and Theories of Dark Matter Halos 4. Supermassive and Intermediate-Mass Black Holes	
JWST Programs	JWST/NIRSpec (36 hr) & NIRCam parallel (27.5 hr)	Principal Investigator (PI) PID 12340
	JWST/NIRCam (1000 hr) pure parallel	Co-Investigator (Co-I) PID 10790
	JWST/NIRSpec (97.4 hr) & NIRCam parallel (75.6 hr)	(Co-I) PID 10149
	JWST/NIRSpec IFU (26.7 hr)	(Co-I) PID 10311
	JWST/NIRSpec (71.7 hr)	(Co-I) PID 8018
	JWST/MIRI LRS (62.8 hr)	(Co-I) PID 8544
	JWST/NIRCam WFSS (12.2 hr)	(Co-I) PID 7336
Honor	Outstanding Graduate, Beijing City & Peking University	May 2023
	First prize, Lin-bridge Scholarship for Astronomy Research	Sep 2022
	First prize, Xingcheng Physics Forum, Peking University	May 2022
	First prize, Mathematics Competition for College Students, Beijing	Dec 2020
	Excellent Undergraduate Research, Peking University	Dec 2022
	Excellent Student Leader, Peking University	Oct 2021
Professional Service	Member, The JWST Advanced Deep Extragalactic Survey (JADES)	2023 – present
	Organizer, Harvard Astronomy Student-Faculty Forum	2024 – present
	Student representative on Harvard Griffin GSAS Student Council	2024 – present
	Student representative on Harvard Astronomy Student Faculty Council	2024 – 2025
	Academic chair, Student Council of School of Physics, Peking University	2020 – 2021
Community Service	Volunteer in Science Festival: <i>Cambridge Explore the Universe</i>	2024, 2025
	Mentor in Directed Reading Program on Cosmology, Harvard Physics Dept.	2026
	Founder of a weekly peer-discussion astronomy seminar (since sophomore)	2021–present
	Bicycle mechanic and cyclist in a 900 km 20-day team cycling expedition	2020
Talks & Posters	Galaxy Discussion Group, University of Cambridge, UK	May 2026
	<i>Talk: A Prominent Overdensity Candidate at $z \approx 10.5$: Galaxy Formation and Clustering in the First 500 Myr</i>	
	Group Meeting, University of Cambridge, UK	May 2026
	<i>Talk: Measuring Dynamical Mass of Dark Matter Halos at $z \approx 7$ from Galaxy Pair Velocities</i>	
	Harvard Institute for Theory and Computation (ITC) Luncheon	Mar 2026
	<i>Talk: A Prominent Galaxy Overdensity Candidate within the First 500 Myr.</i>	
	The 247th AAS meeting, Phoenix, Arizona	Jan 2026
	<i>Talk: JADES-GS-z14-1: A Compact, Faint Galaxy at $z \approx 14$ with Weak Metal Lines from Extremely Deep JWST MIRI, NIRCam, and NIRSpec Observations.</i>	
	JADES Collaboration Meeting, Cambridge, Massachusetts	Jun 2025

Talk: Weak Metal Emission Lines of JADES-GS-z14-1 from Extremely Deep JWST MIRI, NIR-Cam, NIRSpec Observations.

First Galaxies, Oxford, UK

Apr 2025

Flash talk: Stellar Continuum and Nebular Emission of JADES-GS-z14-1 from JWST MIRI/F770W Observations.

JADES Collaboration Meeting, Santa Cruz, California

Jan 2025

Talk: Stellar Continuum and Nebular Emission from JADES-GS-z14-1.

Talk: Wisp Subtraction in JWST NIRCам with the Non-negative Matrix Factorization Algorithm.

JADES Collaboration Meeting, Copenhagen, Denmark

Jun 2024

Talk: MIRI Flux of JADES-GS-z14-0 From Individual Exposures Fitting

PKU-KIAA Seminar, Peking University, China

Jul 2024

Talk: Constraining the Abundance of Intermediate-mass Black Holes from Quasar Microlensing.

Yellow Mountain Guoshoujing Annual Conference, Anhui, China

May 2023

Talk: The Elusive Population of Disk Galaxies with Double Radio Lobes

East Asia AGN Workshop, Chongqing, China

Oct 2021

Poster: AGN Identification from Galaxy 2D Light Profile Decomposition

Selected Press *CfA Press Release* (2024)

Coverage “CfA Astronomers Help Find Most Distant Galaxy Using James Webb Space Telescope”

Sky & Telescope Magazine (2023)

“Unearthing Galactic Gems”

Publication **Zihao Wu**, Daniel J. Eisenstein, Benjamin D. Johnson, Kevin Hainline, *et al.*

“JADES: A Prominent Galaxy Overdensity Candidate within the First 500 Myr”

arXiv e-prints, arXiv:2601.15960 (2026)

Zihao Wu, Benjamin D. Johnson, Daniel J. Eisenstein, Phillip Cargile, *et al.*

“JWST Advanced Deep Extragalactic Survey (JADES) Data Release 5: Wisp Subtraction with the Non-negative Matrix Factorization Algorithm”

arXiv e-prints, arXiv:2601.15958 (2026)

Zihao Wu, Daniel J. Eisenstein, Benjamin D. Johnson, Peter Jakobsen, *et al.*

“JADES-GS-z14-1: A Compact, Faint Galaxy at $z \approx 14$ with Weak Metal Lines from Extremely Deep JWST MIRI, NIRCам, and NIRSpec Observations”

The Astrophysical Journal, 992, 212 (2026)

Zihao Wu, Luis C. Ho

“Detecting Intermediate-mass Black Holes Using Quasar Microlensing”

The Astrophysical Journal, 985, 2 (2025)

Zihao Wu, Luis C. Ho, Ming-Yang Zhuang

“An Elusive Population of Massive Disk Galaxies Hosting Double-lobed Radio-loud AGNs”

The Astrophysical Journal 941, 95 (2022)

P. Rinaldi, G. Rieke, **Z. Wu**, *et al.*

“Beyond the Dot: an LRD-like nucleus at the Heart of an IR-Bright Galaxy and its implications for high-redshift LRDs”

arXiv e-prints, arXiv:2507.17738 (2025)

J. Helton, G. Rieke, S. Alberts, **Z. Wu**, D. Eisenstein, *et al.*

“JWST/MIRI photometric detection at $7.7 \mu\text{m}$ of the stellar continuum and nebular emission in a galaxy at $z > 14$ ”

Nature Astronomy, 1-12 (2025)

F. Sun, D. J. Eisenstein, F. DEugenio, K. Hainline, *et al.*

“JADES: Discovery of Large Reservoirs of Small Dust Grains in the Circumgalactic Medium of Massive Galaxies at $z \sim 3.5$ through Deep JWST/NIRCam Imaging and Grism Spectroscopy”

arXiv e-prints, arXiv:2601.15961 (2026)

K. N. Hainline, D. J. Eisenstein, L. Whitler, B. Robertson, *et al.*

“JWST Advanced Deep Extragalactic Survey (JADES) Data Release 5: Photometrically Selected Galaxy Candidates at $z > 8$ ”

arXiv e-prints, arXiv:2601.15959 (2026)

C. Carreira, B. E. Robertson, A. L. Danhaive, Z. Ji, *et al.*

“JWST Advanced Deep Extragalactic Survey (JADES) Data Release 5: Catalogs of inferred morphological properties of galaxies from JWST/NIRCam imaging in GOODS-N and GOODS-S”

arXiv e-prints, arXiv:2601.15957 (2026)

B. E. Robertson, B. D. Johnson, S. Tacchella, D. J. Eisenstein, *et al.*

“JWST Advanced Deep Extragalactic Survey (JADES) Data Release 5: Photometric Catalog”

arXiv e-prints, arXiv:2601.15956 (2026)

S. Alberts, D. J. Eisenstein, A. J. Bunker, E. Curtis-Lake, *et al.*

“JWST Advanced Deep Extragalactic Survey (JADES) Data Release 5: MIRI Coordinated Parallels in GOODS-S and GOODS-N”

arXiv e-prints, arXiv:2601.15955 (2026)

B. D. Johnson, B. E. Robertson, D. J. Eisenstein, S. Tacchella, *et al.*

“JWST Advanced Deep Extragalactic Survey (JADES) Data Release 5: NIRCam Imaging in GOODS-S and GOODS-N”

arXiv e-prints, arXiv:2601.15954 (2026)

J. M. Helton, J. E. Morrison, K. N. Hainline, F. DEugenio, *et al.*

“Ionizing Photon Production Efficiencies and Chemical Abundances at Cosmic Dawn Revealed by Ultra-Deep Rest-Frame Optical Spectroscopy of JADES-GS-z14-0”

arXiv e-prints, arXiv:2512.19695 (2025)

Z. Zhang, M. Li, M. Oguri, X. Lin, *et al.*

“Little red dot variability over a century reveals black hole envelope via a giant Einstein cross”

arXiv e-prints, arXiv:2512.05180 (2025)

A. L. Danhaive, S. Tacchella, A. J. Bunker, E. Curtis-Lake, *et al.*

“The dark side of early galaxies: geko uncovers dark-matter fractions at $z \sim 4-6$ ”

arXiv e-prints, arXiv:2510.14779 (2025)

J. A. A. Trussler, A. J. Cameron, D. J. Eisenstein, H. Katz, *et al.*

“Cloudy with a chance of starshine: Possible photometric signatures of nebular-dominated emission in $1.5 < z < 8.5$ JADES galaxies”

arXiv e-prints, arXiv:2510.12622 (2025)

F. D’Eugenio, E. J. Nelson, D. J. Eisenstein, R. Maiolino, *et al.*

“JADES Dark Horse: demonstrating high-multiplex observations with JWST/NIRSpec dense-shutter spectroscopy in the JADES Origins Field”

arXiv e-prints, arXiv:2510.11626 (2025)

A. L. Danhaive, S. Tacchella, W. McClymont, B. Robertson, *et al.*

“Beyond the stars: Linking H α sizes, kinematics, and star formation in galaxies at $z \sim 4\text{--}6$ with JWST grism surveys and **geko**”
arXiv e-prints, arXiv:2510.06315 (2025)

Y. Zhu, M. J. Rieke, Z. Ji, A. J. Bunker, *et al.*
“Clump-like Structures in High-Redshift Galaxies: Mass Scaling and Radial Trends from JADES”
arXiv e-prints, arXiv:2601.15965 (2026)

P. Rinaldi, P. Prez-Gonzalez, G. Rieke, *et al.*
“Deciphering the Nature of Virgil: An Obscured AGN Lurking Within an Apparently Normal Lyman- Emitter During Cosmic Reionization”
arXiv e-prints, arXiv:2504.01852 (2025)

J. Witstok, R. Smit, W. Baker, P. Rinaldi, *et al.*
“On the origins of oxygen: ALMA and JWST characterize the multi-phase, metal-enriched, star-bursting medium within a ‘normal’ $z > 11$ galaxy”
arXiv e-prints, arXiv:2507.22888 (2025)